

Table S1. Feature catalog

Base name	Description	Timeframes available	Normalisation / bounds
stoch_rsi_k	Stochastic RSI %K	15m, 4h, 1d	0–100
stoch_rsi_d	Stochastic RSI %D	15m, 4h, 1d	0–100
rsi	Relative Strength Index	15m, 4h, 1d	0–100
bb_position	Bollinger Band position	15m, 4h, 1d	0 = lower band, 1 = upper band
adx	Average Directional Index	15m, 4h, 1d	0–100
ema_diff	EMA fast - EMA slow, normalised by price	15m, 4h, 1d	Percentage of price
trend	EMA trend direction	15m, 4h, 1d	0 = down, 1 = up
returns_1	1-period percentage return	15m, 4h, 1d	Percentage
atr_pct	Average True Range normalised by price	15m, 4h, 1d	Percentage of price
macd_pct	MACD line normalised by price	15m, 4h, 1d	Percentage of price
macd_signal_pct	MACD signal line normalised by price	15m, 4h, 1d	Percentage of price
macd_hist_pct	MACD histogram normalised by price	15m, 4h, 1d	Percentage of price
bb_width_pct	Bollinger Band width normalised by price	15m, 4h, 1d	Percentage of price
volume_norm	Volume normalised by 20-period moving average	15m only	Ratio
returns_5	5-period percentage return	15m only	Percentage

Base name	Description	Timeframes available	Normalisation / bounds
<code>dist_sma200_pct</code>	Distance to SMA200 normalised by price	1d only	Percentage of price
<code>btc_trend</code>	BTC above/below EMA50	1d only	0 = below, 1 = above
<code>atr_regime</code>	Current ATR / 50-period moving average of ATR	15m, 4h, 1d	Volatility ratio

Note: The catalog is enforced by the validator. Each base feature is expanded across its allowed timeframes, yielding concrete column names such as `rsi_15m`, `adx_4h`, or `btc_trend_1d`. Absolute prices and unnormalised quantities (e.g., `close`, `ema_fast`, `atr`) are forbidden.